Engineering Geology 355 (2025) 108252

Contents lists available at ScienceDirect

Engineering Geology

journal homepage: www.elsevier.com/locate/enggeo

Technical note

Physics-based and data-driven long-term evolution of a landslide: From inversion to prediction

Wenjie Du [a,b], Xiaodong Fu [a,b,*], Qian Sheng [a,b], Jian Chen [a,b], Yongqiang Zhou [a,b], Shaojie Zheng [c]

[a] State Key Laboratory of Geomechanics and Geotechnical Engineering Safety, Institute of Rock and Soil Mechanics, Chinese Academy of Sciences, Wuhan 430071, China
[b] School of Engineering Science, University of Chinese Academy of Sciences, Beijing 100049, China
[c] School of Civil Engineering, Chang'an University, Xi'an, Shaanxi 710064, China

ARTICLE INFO

Keywords:
Material point method
Long-term geological disaster evolution
Digitization
Physically driven
Inversion and prediction

ABSTRACT

Leveraging digital technology to improve public service capabilities in disaster prevention and mitigation is a key focus in geological hazard research. This research expands the utilization of numerical simulations for geological disaster forecasting, including studies on current-state inversion, future prediction techniques, and data integration. A geological hazard simulation framework endowed with long-term predictive capabilities was developed using the Material Point Method (MPM), with sensing data of the prototype system state and digital twin models constitute the core data components. A disaster-scenario twin model was established by extracting key elements from disaster scenarios, serving as the data storage medium for digital simulations. State sensing of the disaster prototype scenario was used to conduct system state inversion and optimize the selection of physico-mechanical parameters. The twin model was then updated using physico-mechanical models of geotechnical materials to forecast the future development of the disaster. Taking the Taihe landslide as a case study, the prototype system state was derived from monitoring data, along with the time-dependent characteristics of strength degradation of geotechnical materials. Inversion of disaster state and predictive simulations were performed using the twin model driven by data and physical principles. The landslide deformation direction, range, and local collapse locations between the twin model and the disaster prototype were compared to further achieve the constraints and optimization of the input parameters of the twin model. Building on this, long-term stability simulations for the landslide were performed, integrating the strength evolution of geotechnical materials to incorporate time-span characteristics in the landslide twin model for long-term behavior. The evolution of future disasters of the Taihe landslide over the next decade was forecasted. A disaster-scenario digital twin model, extending traditional numerical simulation in disaster fields, was developed. By integrating data and physics-driven approaches, it creates a simulation framework with temporal generalization capability. This advances the utilization of digital technologies in engineering geology and improves the accuracy of disaster prediction.

1. Introduction

Employing digital technology to improve public services for disaster prevention and mitigation, as well as to enhance efficiency, is a prominent focus for research into geological hazards. The strengths of digital technology in geological hazards lie in two key areas: state sensing and predictive simulation of disasters. State sensing entails the deployment of diverse sensors to detect and gather information pertinent to landslides, such as rainfall, groundwater levels, moisture contents, and deformation data, to reconstruct the current deformation state of the landslide prototype system and evaluate its stability. Non-contact exploration technologies (Frattini et al., 2018), such as synthetic aperture radar, transient electromagnetic methods, and unmanned aerial vehicles (UAVs), provide data on terrain, surface textures, and geological structures and enable the monitoring of regional-scale multi-temporal surface deformation (Bearzot et al., 2022). Even with their initial

* Corresponding author at: State Key Laboratory of Geomechanics and Geotechnical Engineering Safety, Institute of Rock and Soil Mechanics, Chinese Academy of Sciences, Wuhan 430071, China.
E-mail address: xdfu@whrsm.ac.cn (X. Fu).

https://doi.org/10.1016/j.enggeo.2025.108252
Received 7 February 2025; Received in revised form 9 July 2025; Accepted 12 July 2025
Available online 13 July 2025
0013-7952/© 2025 Published by Elsevier B.V.

limitations, aerial and satellite imagery provided a vital instrument for regional-scale analysis, identifying areas with a historical propensity for instability and highlighting geological structures that might predispose slopes to failure. Equipped with high-resolution optical cameras, UAVs can capture detailed imagery of landslide areas, thus enabling the precise mapping of landslide boundaries and morphological characteristics, and the identification of subtle changes in surface conditions over time. This capability is invaluable to understanding the spatial extent and characteristics of landslides, as well as for detecting early signs of activity or assessing post-failure scenarios (Casagli et al., 2010; Casagli et al., 2023).

Predictive simulation encompasses the assessment of landslide susceptibility and stability using techniques such as geostatistical methods and physico-mechanical approaches (Gong et al., 2024; Reichenbach et al., 2018; Merghadi et al., 2020). Geostatistical methods begin with an examination of the spatial autocorrelation of landslide occurrences, summarizing disaster-inducing factors from numerous landslide samples to evaluate the susceptibility of a region to geological hazards. However, the lack of a deterministic analysis of the physico-mechanical mechanisms underlying these phenomena often leads to predictive bias induced by an over-reliance on data. Physico-mechanical approaches usually involve solving disaster models through governing equations, enabling inversion of the current stability state and projection of future evolution trends. Physico-mechanical evaluation methods are primarily applied to the analysis of localized single-point landslide, with stability assessment results often aligning more closely with physico-mechanical principles. However, their application in disaster forecasting and early warning remains limited. The underlying reason for this is that the development of simulation algorithms for movement and accumulation of post-initiation disasters in the geological hazard field tends to prioritize phenomenological approximations. Emphasis on approximating the final deposition form often overlooks critical details of the disaster evolution process, diminishing the reliability of predictions for landslide initiation and dynamics. Moreover, simulations and predictive techniques exhibit limited long-term predictive capabilities. The integration of most algorithms and models is tailored to specific disaster types or specific working conditions. The temporal concepts in calculations are often ambiguous, hindering the connection between numerical simulation outcomes and disaster forecasting efforts.

Additionally, with the widespread adoption of high spatio-temporal resolution mapping and monitoring data collection technologies and the rapid advancement of emerging computational sciences, data storage formats and representations have become increasingly intricate. The fusion and processing of multi-source heterogeneous data pose a substantial challenge for the practical implementation of digital applications in the realm of geological disasters (Li et al., 2022). Data serve as the cornerstone of digital technology. The twin model acts as a storage platform for predictive simulation data, while diverse external data sources drive the dynamic inversion and operation of the twin model. The integration and interaction between them are essential for realizing dynamic simulation, monitoring, and analysis of physical entities within a digital environment.

The rapid advancements in digital technologies such as the Internet of Things (IoT), big data, cloud computing, and artificial intelligence have led to the emergence of the transformative paradigm of “digital twinning.” Digital twins utilize IoT-based monitoring devices to collect data on field conditions, mechanical responses, and geological structures in engineering geomechanics. Using advanced data analytics and artificial intelligence, real-time inverse analysis and feature extraction are conducted, enabling the continuous spatio-temporal evolution of geomechanical parameters. Integrating cloud computing, multi-field continuous-discontinuous computations, and advanced technologies such as deep learning, digital twins' advance stability analysis, risk assessment, and insights into complex geomechanical mechanisms. Simultaneously, advancements in monitoring technologies — including interferometric synthetic aperture radar, microseismic monitoring, fiber-optic-based full deformation, laser scanning, and computed tomography — along with innovative macro and micro-scale numerical algorithms, including the finite-discrete element method, smoothed particle hydrodynamics (SPH), material point method (MPM), particle finite element method (PFEM), peridynamics (PD), phase field method (PF), and numerical manifold method (NMM), have opened additional avenues for advancing this paradigm (Conte et al., 2019; Conte et al., 2020; Yerro et al., 2019; Soga et al., 2016; Zhang et al., 2023; Dimililer et al., 2020; Giri et al., 2019; Du et al., 2023). The widespread adoption of machine learning, data-driven computation, and cloud computing further facilitates new pathways for landslide prediction and forecasting using digital twins. Embedding physical constitutive and governing equations into neural networks (e.g., PINNs) is a hot topic at the nexus of machine learning and computational mechanics. Using PDE residuals as part of the loss function makes neural network outputs follow physical laws. In data-scarce scenarios, physical constraints boost model generalization and support multi-equation coupling modeling in complex systems (Wang et al., 2024).

This research focuses on the application of extended numerical simulations grounded in the digital twin concept for forecasting geological disasters. Taking the Taihe landslide as a case study, it explores data-driven inversion of current disaster conditions, physics-driven prediction of disaster evolution, and data fusion. A simulation framework with time generalization capability is constructed adopting the MPM. Coupled with the evolution of properties of geotechnical materials, multi-perspective and multi-dimensional scenario analyses can reveal the mechanisms underpinning the initiation of landslide instability and predict its future evolution.

2. Material and methods

2.1. The Material Point Method

In Material Point Method (MPM), the continuum body is discretized into material points (Lagrangian) and a background computational grid (Eulerian), which is used to solve the governing equations. The method, first proposed by Sulsky et al. (1994), utilizes the Dirac function to map the physical quantity f_p (such as density, stress, and acceleration) from the MPs to the grid.

$$f(x) = \sum_p f_p \delta(x - x_p)$$

By using the background grid shape functions N_{Ip} to approximate the virtual displacement and then reformulating the virtual work equation, the equations of motion for the background grid nodes are obtained.

$$\dot{p}_I = f_I^{\text{int}} + f_I^{\text{ext}}$$

The expressions for the nodal internal and external forces are, respectively:

$$f_I^{\text{int}} = -\sum_{p=1}^{n_p} \frac{m_p}{\rho_p} \sigma_p \cdot \nabla N_{Ip}$$

$$f_I^{\text{ext}} = \sum_{p=1}^{n_p} N_{Ip} \bar{t} h^{-1} \frac{m_p}{\rho_p} + \sum_{p=1}^{n_p} m_p N_{Ip} b$$

In the equation,Irepresents the grid node, b denotes the body force, $\bar{t}$ represents the boundary surface force, and h is the hypothetical boundary layer thickness.

In the standard Material Point Method (MPM), when a MP crosses a grid, the discontinuity in the derivatives of the shape functions causes grid-crossing errors. Considering the large-scale computational accuracy and stability for practical engineering applications, and aiming for larger time steps, this study adopts the Generalized Interpolation Material Point Method (GIMP) with USF format for calculations, and

utilizes the leapfrog Verlet central difference algorithm for explicit solution computation.

Bardenhagen et al. (Bardenhagen and Kober, 2004) developed the Generalized Interpolation Material Point Method (GIMP) based on the Petrov-Galerkin method, By using an asymmetric test function-shape function pair, i.e., the background grid function $N_I(x)$ and the particle characteristic function $\chi_p(x)$, the weak form integration process is reconstructed, with the particle's domain serving as the continuity support for the shape function, which effectively suppresses grid-crossing errors. In GIMP, the characteristic function $\chi_p(x)$ of the MP is selected as the trial function to approximate the physical quantities of the system f_p in the format of

$$f(x) = \sum_p f_p \chi_p(x)$$

Accordingly, the expressions for the nodal internal and external forces are updated to:

$$f_I^{\text{int}} = -\sum_{p=1}^{n_p} \frac{m_p}{\rho_p} \sigma_p \cdot \nabla S_{Ip}$$

$$f_I^{\text{ext}} = \sum_{\Gamma_t} N_{Ip} \bar{t} d\Gamma + \sum_{p=1}^{n_p} m_p S_{Ip} b$$

Here, Γ is the given boundar, and the shape function S_{Ip} in GIMP now denotes the mapping function between the background grid and the MPs in GIMP. Compared to the standard MPM, GIMP enhances the continuity of the shape function, reducing numerical noise across grids.

After completing the mapping of information between the material points and the background grid, strain and stress updates are performed, the background grid is used to calculate the deformation gradient (Bardenhagen, 2002; Nairn, 2003), and the stress state of the MPs σ_p is subsequently updated based on the constitutive equations. On this basis, the momentum equation is solved to update the velocity changes of the nodes and mapped back to the material points. The material points carry the updated physical information into the next time step calculation, for more details about MPM, please reference to Fern et al. (2019) and Du et al. (2025).

2.2. Analysis step inheritance algorithm

Time-dependent characteristics are a fundamental mechanical property of geotechnical materials, however, conventional landslide stability analysis mainly focuses on the elasto-plasticity of these materials, overlooking the temporal evolution of their stability state. The long-term stability of landslides, which is influenced by both internal and external factors, is closely related to variables such as fluctuations in the groundwater level and rainfall intensity. Evaluating long-term stability of landslides generally involves examining both external factors and the time-dependent characteristics of the mechanical properties of the material (Federico et al., 2011; Alonso, 2021).

Analysis of landslide long-term stability typically requires calculations over periods measured in years. Due to issues such as time-step size and computational efficiency, explicit analysis methods necessitate a considerable amount of time to calculate the long-term stability. The primary factors controlling landslide long-term stability include the degradation of strength parameters and the topographical changes caused by cumulative deformation. This study selects key points in the long-term deformation process for analysis, breaking down the calculation of long-term stability into a series of analytical steps. Each step terminates when convergence is reached or motion becomes stable, with positions, stress states, displacements, velocities, volumes, and elasto-plastic states of material points (MPs) from the previous analysis step output and used as input for the current analysis step. During the initial geostress analysis step, MP displacement, velocity, and elasto-plastic states are reset to zero. The flowchart through the analytical step-inheritance algorithm is shown in Fig. 1.

2.3. Extraction of disaster-scenario elements and state sensing

The excavation in southern region of the Taihe Mine exposes a complex geological setting (Fig. 2). From oldest to youngest, the formations include Hercynian gabbro and diabase revealed through mining, Quaternary Middle-Lower Pleistocene till (Q_{1-2}^{fgl}), Quaternary Holocene landslide deposits (Q_4^{del}), Quaternary Holocene colluvium (Q_4^{dl}), Quaternary Holocene alluvium (Q_4^{al+pl}), and newly formed artificial deposits (Q_4^{ml}). Observations from excavation, drilling, and trench exploration indicate that the ancient sliding surface, now appearing as a shear outlet modified by terrain shifts, is predominantly aligned with the excavation-exposed bedrock-deposit interface from west to east. The interface outcrops higher in the west, sloping lower toward the east, forming a counter-dipping slope. The sliding body primarily comprises Quaternary glacial till debris, with the sliding zone positioned within the soil layer and the bedrock-deposit interface. The excavation activities expose the interface between the bedrock and the Quaternary deposit layers, delineating the northern boundary of the landslide. By the rainy season in 2022, cracks along the landslide boundary exhibited a clear chair-like pattern, and the landslide became fully formed (Fig. 3).

The landslide develops in the central part of the southern pit slope of the open-pit mine: its planform is roughly semicircular, with a maximum longitudinal length of approximately 495 m, an average width of 654 m, an average thickness of 37 m, and a volume of about 1200×10^4 m^3, making it a giant soil landslide. It slides in a northeastward direction at 18–39°, with an average vertical displacement of 145.0 m along the slip direction. The landslide area presents an overall topography with higher elevations in the west and lower ones in the east, with a maximum height difference of 226 m along that direction. From east to west, the landslide area contains 17 steps, each with a height of 12 m and platform widths ranging from 4 to 9 m. The slope angle is approximately 30–45°, and the terrain near the back edge of the landslide transitions from gentle to steep on the mountain side.

According to rainfall statistics from 2016 to 2021 (Fig. 4), the rainfall from May to September constitutes 86.2 % of the total annual rainfall, with the average monthly rainfall from June to September accounting for 77.6 % thereof. The rainfall period is concentrated, with frequent heavy and torrential rain, and a high incidence of continuous and nighttime rainfall. The dry and wet seasons are distinct; most of the local collapses on the pit slopes occur in the rainy season from May to

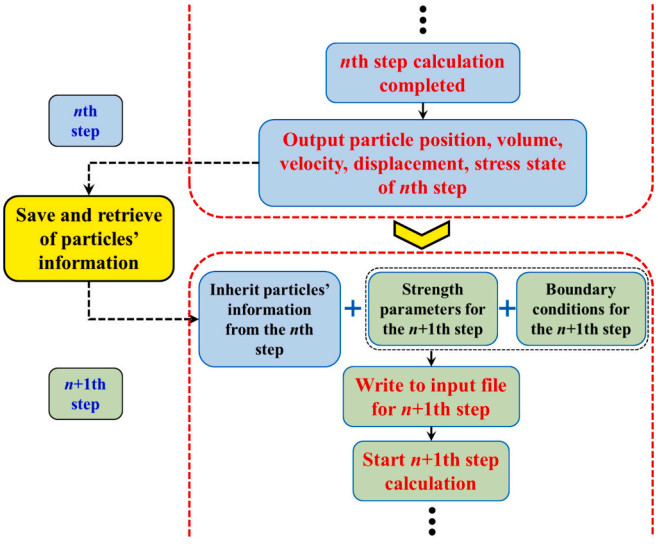


Fig. 1. Flowchart of the Analysis Step Inheritance Algorithm.

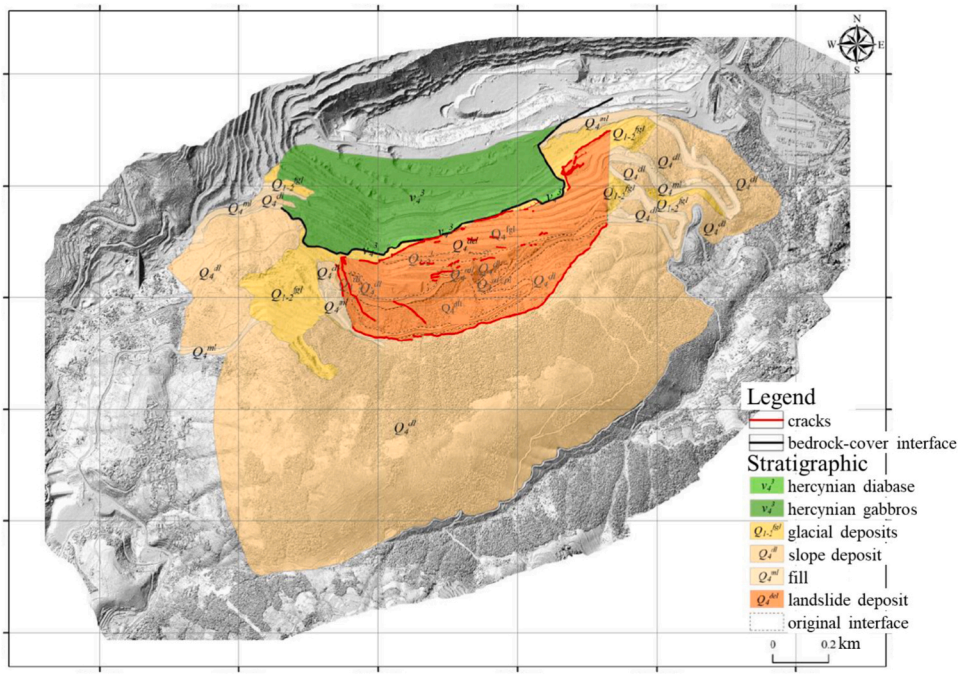


Fig. 2. Distribution of Stratigraphic Lithology.

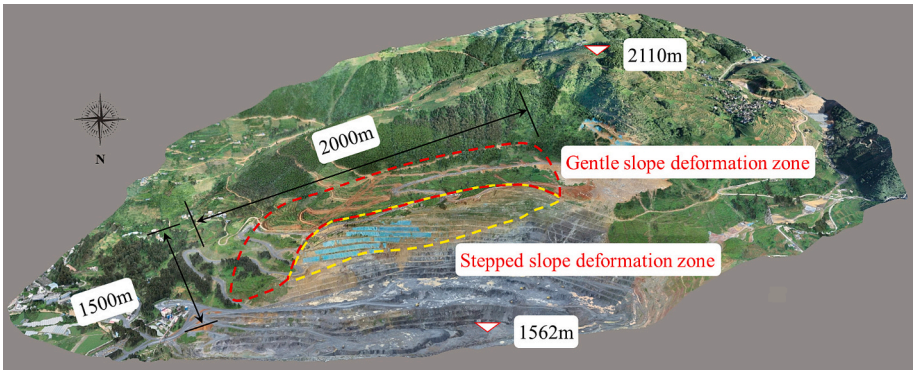


Fig. 3. Overview of the Taihe Landslide.

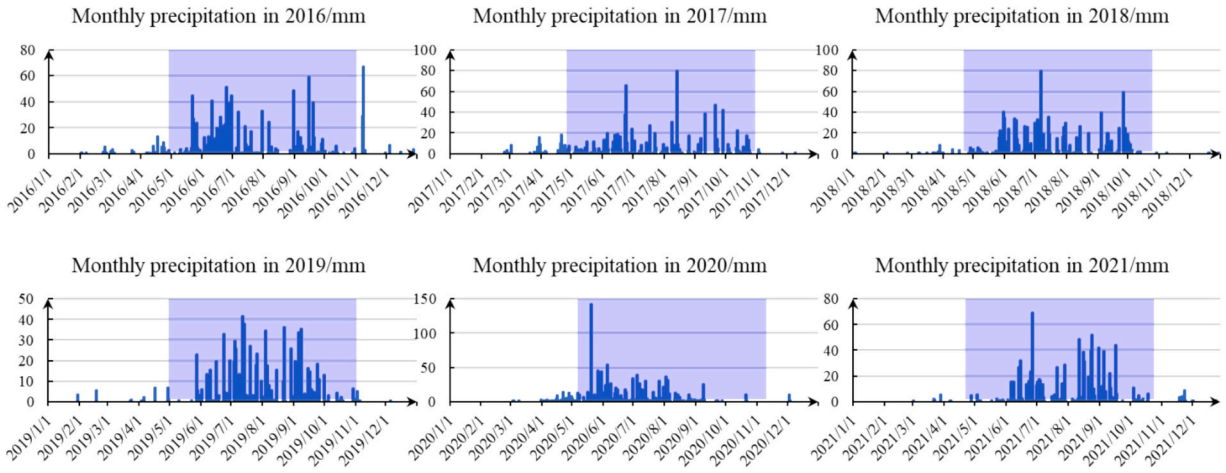


Fig. 4. Precipitation Statistics of the Disaster Site.

September each year.

The digital twin framework for landslides centers on data of the prototype system state, using the twin model as a bridge to facilitate dynamic interactions. This includes prototype system state-driven processes, twin model coupling, and feedback of state information via mutual information exchange. At the heart of the framework is the fusion of multi-source data with real-time updates, aimed at building a digital twin capable of dynamically representing the actual state of a given landslide. Specifically, the system state data of the prototype encompass terrain morphology, geological structure, surface and deep deformation, geotechnical parameters, rainfall, and groundwater. These datasets are obtained via UAVs, electromagnetic surveys, sensors, and global navigation satellite systems (GNSS). Leveraging integrated multi-source data, a 3-d geological model of the landslide twin is built, intuitively representing the current state of stability of the landslide. Building on this, the evolution of properties of the salient geotechnical material is integrated to enable dynamic updates to the deformation and stability states of the landslide.

Taking the construction of Taihe landslide digital twin scenario as an example, the process of geological feature extraction and state sensing is as follows: 1) High-precision surface topography data are acquired utilizing drones equipped with high-resolution cameras or LiDAR, producing a digital elevation model and orthophotos to supply topographic data for the twin model; electromagnetic methods are employed to deduce subsurface rock layer distribution, fault positions, and groundwater distribution, offering data of geological structures for the twin model; 2) Surface and subsurface deformations of the landslide are monitored using GNSS and inclinometers, capturing deformation magnitude, direction, and rate data, which are used to describe the landslide deformation mechanism and stability, enabling the inversion of the current state of the landslide; 3) Laboratory physico-mechanical tests are conducted to determine critical parameters such as elastic modulus and shear strength of the landslide mass: these are integrated with the trend in material properties and strength parameters to facilitate the analysis of landslide deformation and long-term stability predictions. Fig. 5 depicts the complete workflow through geological feature extraction and state sensing in the Taihe landslide digital twin scenario, encompassing all steps from data collection and model development to state feedback.

Combining landslide state data and displacement-rate time series, the displacement increment is approximately 3 mm/d during the dry season, while during the rainy season, the displacement rate ranges from 3 to 10 mm/d. The surface displacement of monitoring points within the landslide is significantly affected by rainfall, and the development of its displacement is strongly correlated with cumulative rainfall. Cumulative displacement and deformation rates are positively correlated with rainfall, with stable increments during the dry season. Since the second half of 2022, the monitoring curve has entered an accelerated deformation phase, with the landslide generally undergoing creep and stable deformation, although it has not yet converged. Radar monitoring results indicate that the deformation control surface of the Taihe landslide is the bedrock-deposit interface, where significant relative deformation occurs near the interface. The surface displacement vector and the deep displacement vector are essentially consistent:

(I) The direction of the displacement vector is predominantly northeastward, ranging from 10° to 40°. The angle of the displacement vector in the middle of the landslide body is larger than that at the rear edge;
(II) Deep displacement monitoring shows that the landslide body is largely intact, with no secondary slip surfaces identified. The consistency in deformation further suggests that the landslide remains cohesive and has not disintegrated, remaining in a creeping state;
(III) The displacement outside the boundary of the landslide is small.

2.4. Construction of the digital twin model

A top-down modeling approach is employed to create a structured 3-d MP model with equal spacings such that $\Delta z = \Delta x = \Delta y$, using the MP spacing as the resolution to extract surface and stratigraphic interface point cloud models (Fig. 6). The stratigraphic interface point cloud (including the glacial deposit/strongly weathered diabase interface, strongly weathered diabase/medium weathered gabbro interface, and sliding surface) is employed to slice the 3-d MP model. A binary tree model cutting algorithm is introduced to create a spatial position index between the MP model and the surface/stratigraphic point cloud data. The 3-d MP model is sliced based on this index, and material types are grouped and defined. The physical model is divided into five groups: the landslide mass, landslide surface, depositional body, strongly weathered

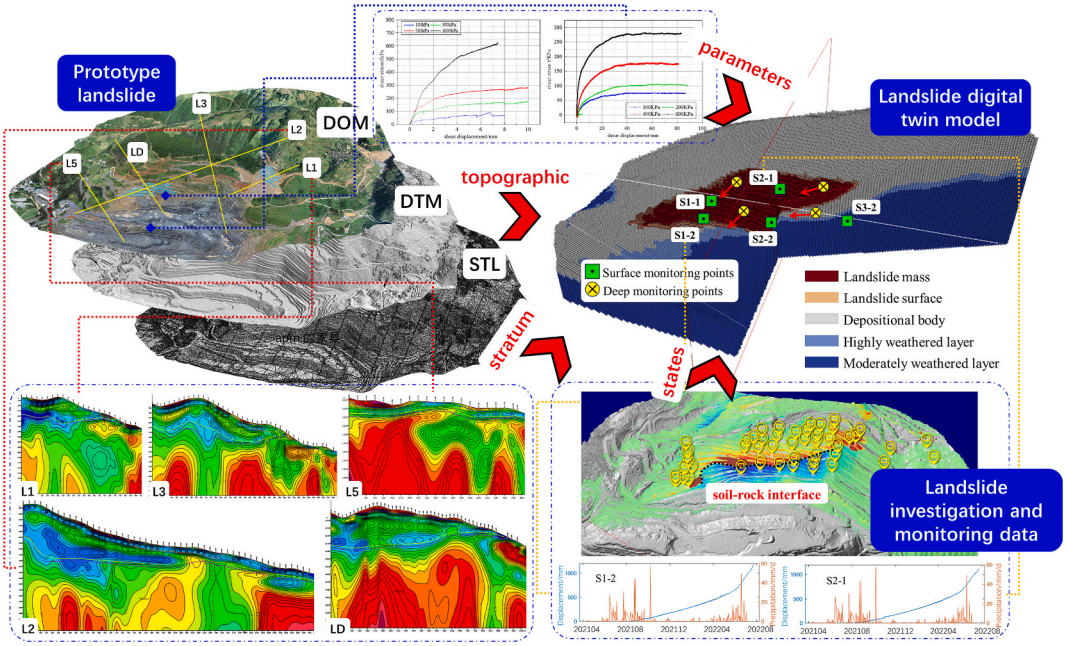


Fig. 5. Schematic Diagram of Geological Element Extraction and State Perception for Disaster Scenario.

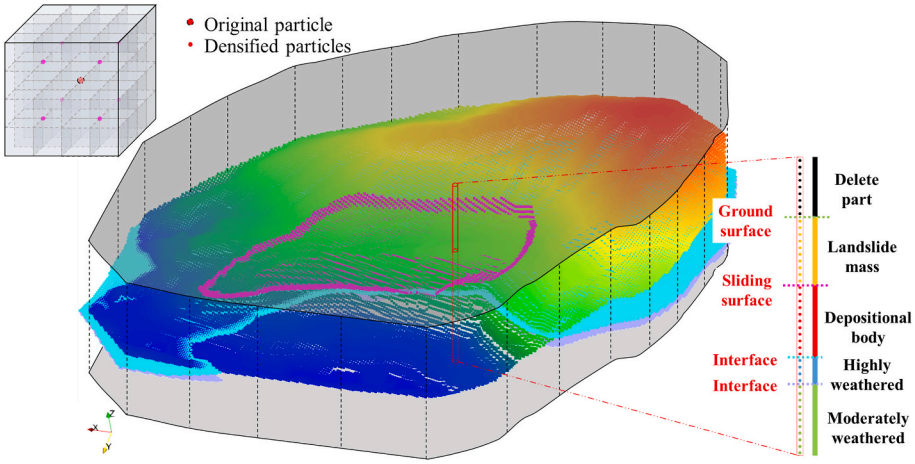
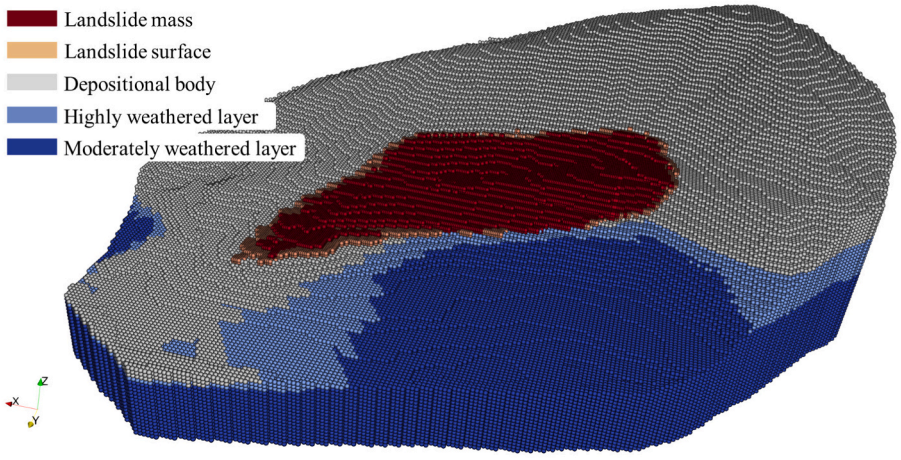
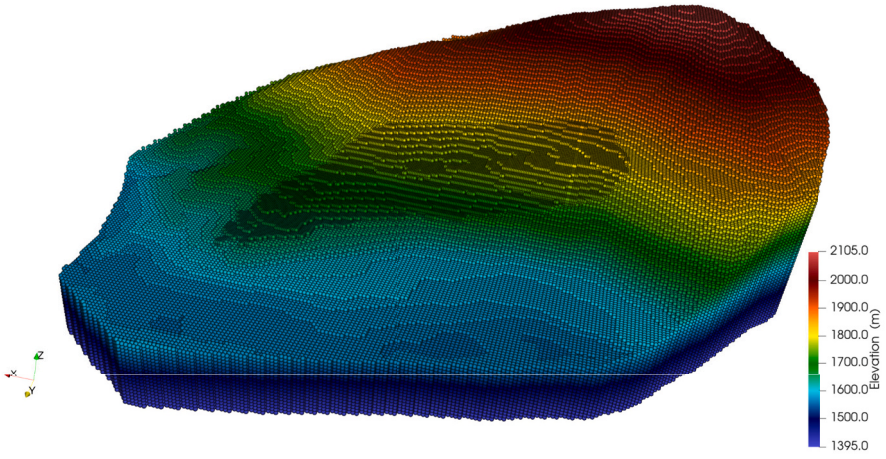


Fig. 6. Schematic Diagram of Cutting the 3D Material Point Geological Model.



(a) Material Grouping



(b) Model Elevation

Fig. 7. Landslide Numerical Calculation Model.

layer, and medium weathered layer (Fig. 7). By synthesizing computational accuracy and efficiency, the geometric configuration of the model is ultimately determined as follows: the model consists of 1,162,841 MPs (with a global spacing of 10 m; locally refined to 5 m for the landslide mass) and a structured background grid (10 m). The volume of the deformed body in the disaster model is 1236×10^4 m^3, which closely

matches the geological survey estimate of 1200 × 10^4 m^3. The background grid employs Cartesian cubic elements, each measuring 10 m × 10 m × 10 m. Each element in the moderately weathered layer, strongly weathered layer, and depositional body contains one MP, whereas each unit cell in the sliding zone and landslide body comprises eight MPs.

3. Digital simulation of the entire disaster evolution process

Accurately mapping the current state of geological disasters is essential for predicting and forecasting their future evolution utilizing the digital twin technology. The former involves an inversion process targeting the physical real-world state, while the latter requires combining the twin model that maps the real-world state with the laws of physico-mechanical evolution to simulate the future progression of the disaster. The process of disaster state inversion begins with a qualitative assessment of the stability control factors influencing the factor of safety of the landslide based on empirical knowledge. This is followed by inverting the landslide safety factor through use of deformation monitoring data and limit equilibrium analysis, optimizing the physico-mechanical parameters of soil. Ultimately, the optimal parameters for the mapped landslide prototype system as input were obtained for subsequent simulation. Subsequently, site surveys and three-dimensional monitoring are utilized to acquire deformation and damage details of the landslide, such as surface deformation, deep deformation, and locations of localized collapse. These, combined with the physico-mechanical parameters derived from qualitative inversion, are adopted to quantify the deformation and damage details of the landslide prototype (displacement values, areas of concentrated deformation, plastic zones). The physico-mechanical parameters of the landslide are further optimized to create a digital twin state corresponding to the current disaster prototype system, ensuring consistency in physico-mechanical parameters, stability states, and deformation and damage details. Building on this, a model update algorithm driven by nonlinear mechanics and evolution laws of geotechnical materials is adopted. This accounts for the time-dependent evolution of soil physico-mechanical parameters under the influence of dry-wet cycles on the southern pit slope of the Taihe Mine. A physical model approximating the landslide deformation mechanism is developed to represent and predict displacement of the landslide, enabling forward simulation and disaster forecasting of its future evolution. Herein, the prototype system refers to the actual Taihe landslide, while the twin model is a 1:1 particle model reconstructed from the landslide prototype system.

3.1. Inversion of prototype state data-driven model state

Deep displacement monitoring of the Taihe landslide shows that the landslide body is largely intact, with no secondary slip surfaces detected. The consistency with monitoring data of surface deformation further confirms that the landslide mass has not disintegrated and remains in a creeping state. The changes in deformation are stable, but the overall system has not yet converged. Under this premise, profiles are extracted from the digital twin model for the inversion of the geotechnical parameters of landslides, considering the three-dimensional effects. Two profiles, A-A′ and B-B′ (Fig. 8), are selected. A comparison with geotechnical survey profiles further corroborates the accuracy of the constructed digital twin model in depicting the geological structure of the landslide prototype system. The parameters of the two profiles of the landslide body are inverted using the Morgenstern-Price method, thereby determining the comprehensive values for the parameters of slip zone soil: an apparent cohesion of 39 kPa and angle of internal shearing resistance of 7°. On this basis, the material parameters for the stratigraphy of the Taihe landslide are obtained, incorporating laboratory physico-mechanical experimental results, and are listed in Table 1.

The key issue in numerical simulations of geological disasters is to determine the initial geostress field of the model, which determines the reasonability of the subsequent numerical results. Initially, the whole model is assigned an elastic constitutive law and run for 10 s under gravity. Based on this, each layer is transformed from an elastic to an elastoplastic model, with actual deformation and strength parameters applied. These are adopted as the initial geostress field for the digital twin model. As shown in Fig. 9, the distribution of initial geostress is generally aligned with typical patterns.

According to the slip zone parameters obtained from limit equilibrium inversion analysis, under the limit equilibrium state of the landslide, as the deformed mass continues to slide, the pit slope reveals the ancient slip surface, with the upper mass exhibiting noticeable external displacement, characterized by shearing features. The simulation results of stepped slope sliding deformation correspond to the actual landslide in terms of deformation state and range (Fig. 9b), with deformation concentrated within the deformed body surrounded by saddle-shaped boundary cracks.

In the direction of deformation, the deformation vector in the gentle-slope sliding deformation zone is slightly tilted toward the north-east (inversion simulation: front part 12°NBE; rear part 39.6°NBE and actual monitoring: front part 13.8°NBE; rear part 41.9°NBE). On the

Table 1
Stratigraphic Material Parameters of the Taihe Landslide.

Stratum	Unit Weight / kN/m^3	Cohesion / kPa	Internal Friction Angle/ °	Elastic Modulus/ MPa	Poisson's Ratio
Landslide mass	23.0	32.0	26.0	120.0	0.33
Landslide surface	22.0	39.0	7.0	16.0	0.38
Depositional body	23.0	200.0	26.0	1200.0	0.3
Highly weathered layer	30.0	5040.0	41.67	6000.0	0.28
Moderately weathered layer	33.7	6700.0	50.57	30,000.0	0.2

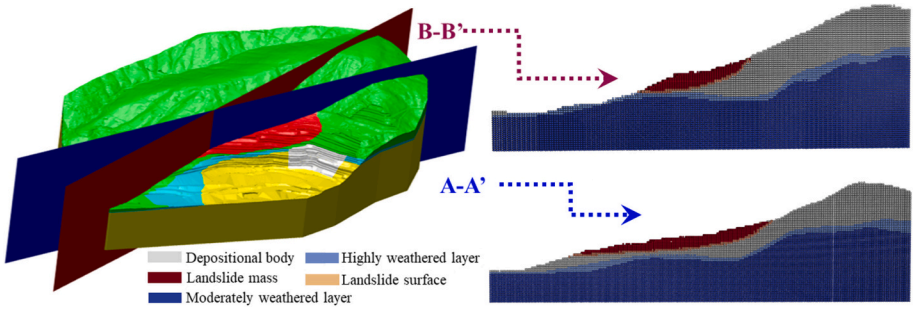


Fig. 8. Profile Extraction of the Twin Model.

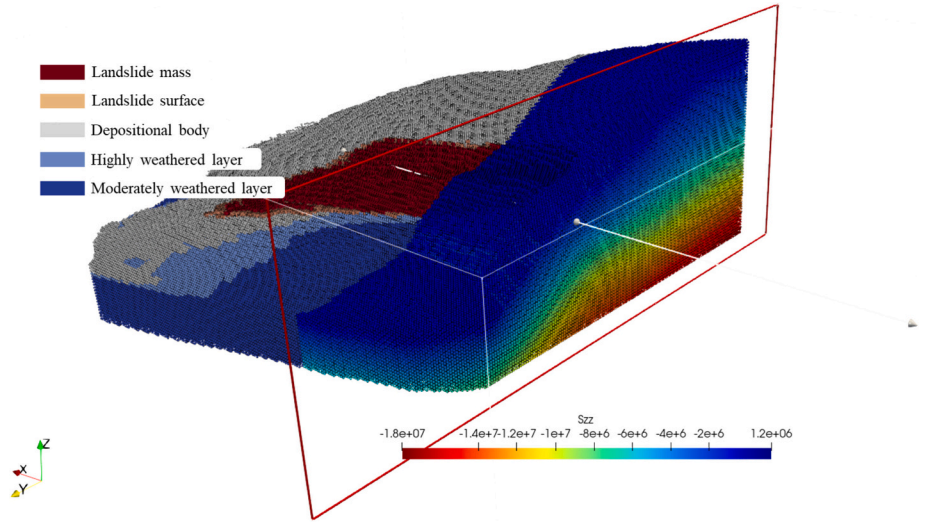


Fig. 9. Initial Equilibrium Analysis Step of the Landslide Model - Distribution of Stress.

eastern side, the front edge of the ancient slip body is blocked by the step corner, causing the deformation to deflect inward toward the mining area. In the stepped slope sliding deformation zone, the deformation shifts inward toward the mining area with a large north-eastward angle. This indicates that the deformation vector direction of the landslide mass is largely consistent with the landslide deformation patterns revealed by surface and deep monitoring (Fig. 10a and c).

Fig. 11 depicts the deep deformation in the digital twin model under the current limit equilibrium state. Points are distributed along the overall sliding direction of the ancient landslide and the sliding directions of two potential instability areas on the eastern and western sides of the southern pit slope. The results show that in the limit equilibrium state, the Taihe landslide undergoes overall deformation along a defined sliding surface, and the two deformation areas on the east and west sides are potential zones of instability. Displacement along the main sliding axis of the ancient landslide increases first and then decreases, while deformation outside the range of landslide deformation is negligible.

The current digital twin model for landslide disasters is in a critical state of instability, with two potential disaster-prone deformation areas located on the eastern and western sides. The overall deformation direction of the landslide is 35° north-east, with deformation on the eastern side being deflected toward the mining area at 59° north-east due to obstruction at the toe of the Taihe landslide. Considering both surface and deep deformation of the landslide, the failure mode involves creep deformation toward the open-pit mine, controlled by the bedrock-deposit interface. The landslide has not yet disintegrated, with the maximum deformation of 0.92 m occurring at the bedrock-deposit interface at the stepped corner on the eastern side, which aligns closely with the location of the current local collapse of the landslide.

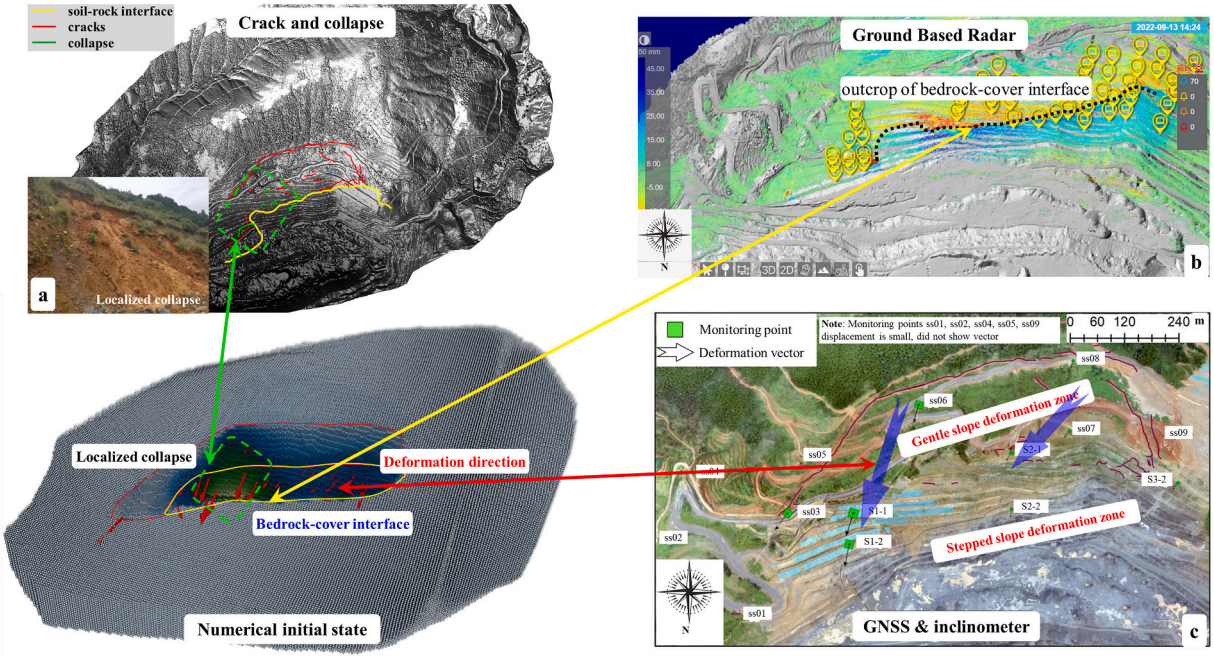
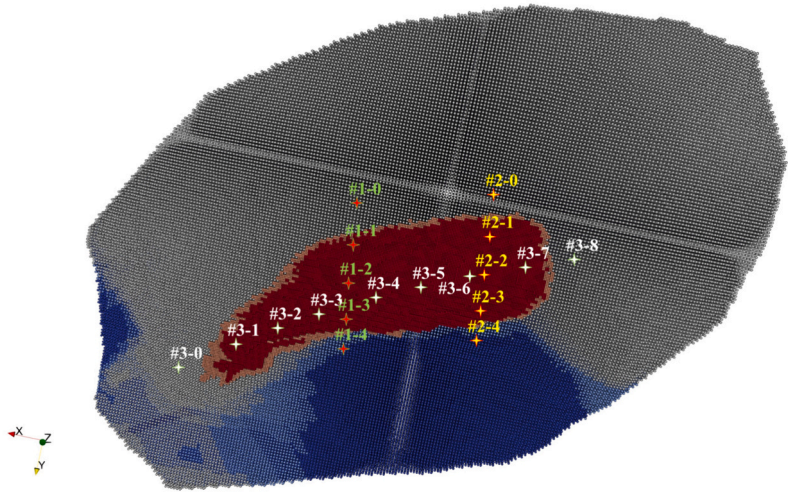
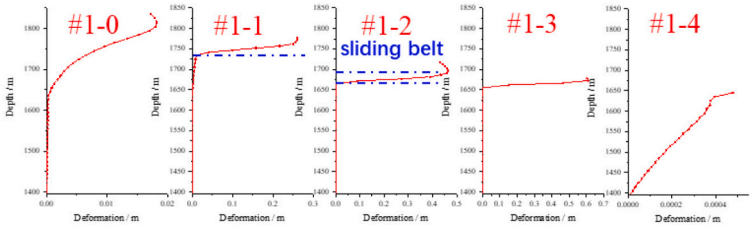


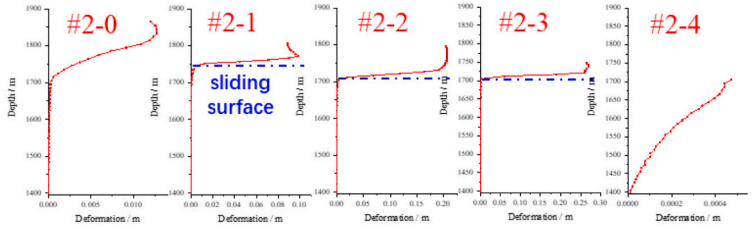
Fig. 10. Comparison of Inverted State and Actual State of the Twin Model.



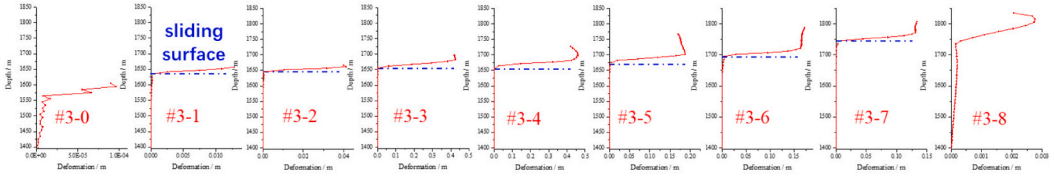
(a) Monitoring Point Layout



(b) #1



(c) #2



(d) #3

Fig. 11. Deep Deformation of the Twin Model.

3.2. Long-term stability simulation driven by the state evolution

3.2.1. Evolution of the prototype state

The landslide area is covered by thick loose Quaternary layers with low permeability and relatively rich shallow groundwater. Groundwater seepage activity as a major factor contributing to landslides and other geological hazards has a significant impact on the stability of the pit slope. The landslide site has an average annual precipitation of 1013 mm, with the rainy season occurring from May to October, accounting for 94 % of the annual average precipitation. Precipitation from November to April of the following year accounts for only 6 % of the annual total. During the rainy season (May to October) each year, continuous rainfall leads to severe seepage on the pit slope, resulting in large areas of water accumulation in the low-lying areas at the base of the pit slope. It is believed that the landslide body, particularly the slip zone above the bedrock-deposit interface, is undergoing repeated wet-dry cycles, which is a key factor governing the future stability evolution of the Taihe landslide.

To characterize the evolution of the mechanical strength of geotechnical materials under wet-dry cycling conditions, over 20 sets of wet-dry cycle experimental data were collected (Deng et al., 2017; Jiang, 2019; Dong et al., 2016; Zhou et al., 2022; Yang, 2021; Dong et al., 2018; Liu et al., 2021; Chang et al., 2022; Zhao et al., 2021; Xue et al., 2022). After normalization, the degradation degree of strength parameters with respect to the number of wet-dry cycles was statistically processed (Fig. 12). The laboratory analysis of soil degradation under wet-dry cycling revealed that when the number of cycles exceeds 10, the apparent cohesion of the material generally decreases by approximately 50 %, and the angle of internal shearing resistance decreases by around 40 %. On this basis, the functional form and value range of the parameter evolution fitting function under long-term wet-dry cycling were determined. An exponential function model, which has clear physical and mathematical significance, was applied to fit the weakening of soil-strength parameters under alternating dry and rainy seasons. The fitted function predicts the accumulation process of deformation and damage resulting from the degradation of strength parameters in the slip zone and landslide body under long-term wet-dry cycling.

The weakening process of soil strength parameters over a 10-year period is described by the exponential function model established earlier (Fig. 13), which demonstrates the degradation pattern of the strength parameters in the slip zone:

$$c = 39.0\left[0.6 + 0.4e^{-0.387N}\right]$$

$$\varphi = 7.0\left[0.8 + 0.2e^{-0.239N}\right]$$

The degradation pattern of strength parameters in the landslide body is given by:

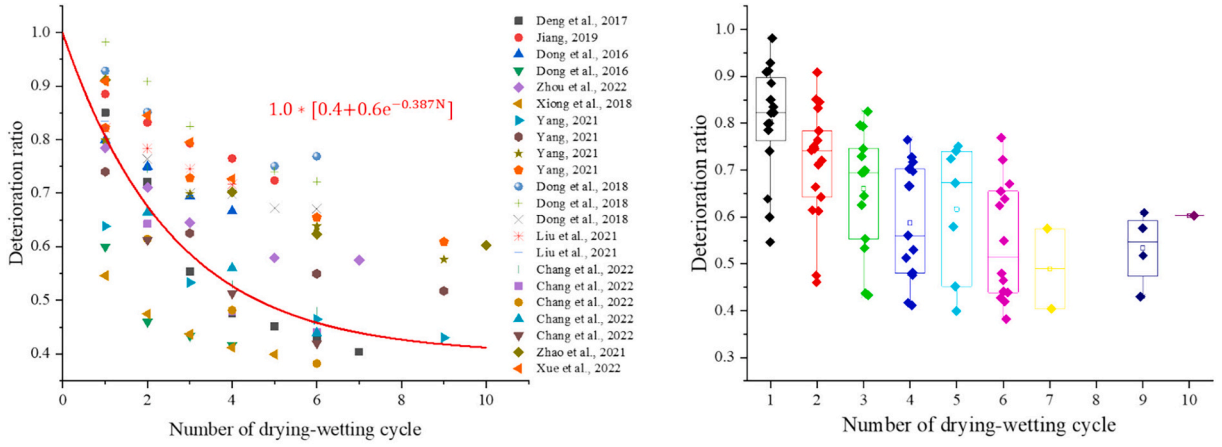
$$c = 32.0\left[0.6 + 0.4e^{-0.387N}\right]$$

$$\varphi = 26.0\left[0.8 + 0.2e^{-0.239N}\right]$$

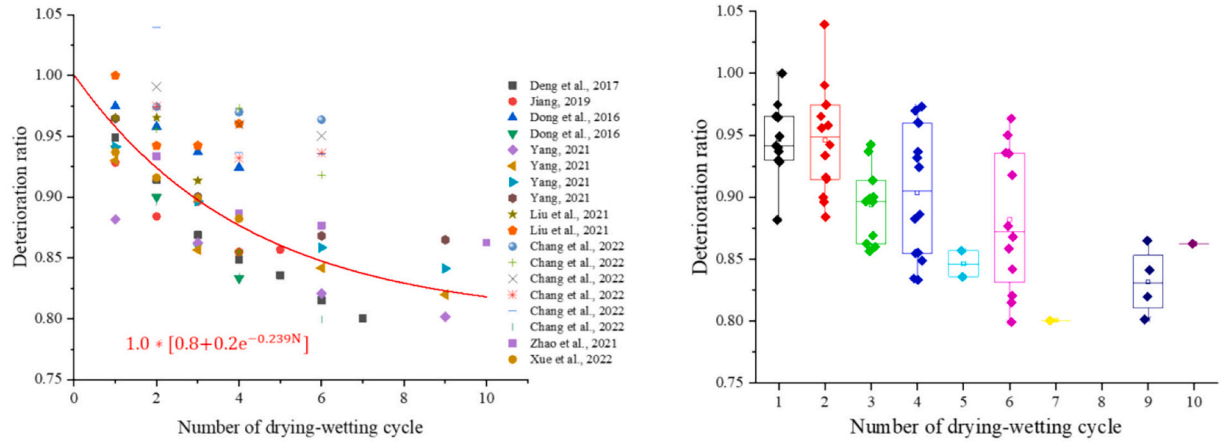
where, N represents the number of degradation cycles, denoting years herein.

3.2.2. *Long-term service performance of Taihe landslide*

Based on the aforementioned long-term cumulative degradation model, it is roughly assumed that one year of alternating dry and rainy seasons corresponds to one wet-dry cycle degradation period. The degradation effect of alternating dry and rainy seasons on the strength of the body of the landslide is considered to assess the instability process and zone of influence of the Taihe landslide under such long-term cumulative degradation. Ten analysis steps for long-term evolution are set up to analyze the instability and destruction process of the Taihe landslide under 10 wet-dry cycle degradation periods over 10 years. The



(a) Cohesion Degradation Law



(b) Internal Friction Angle Degradation Law

Fig. 12. Long-Term Cumulative Degradation Model of Geotechnical Materials.

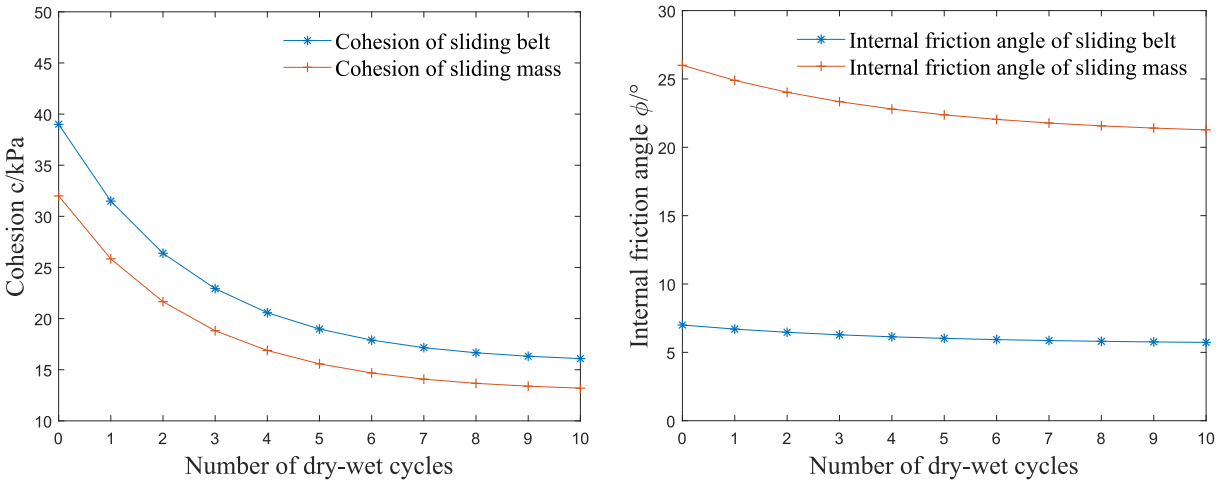
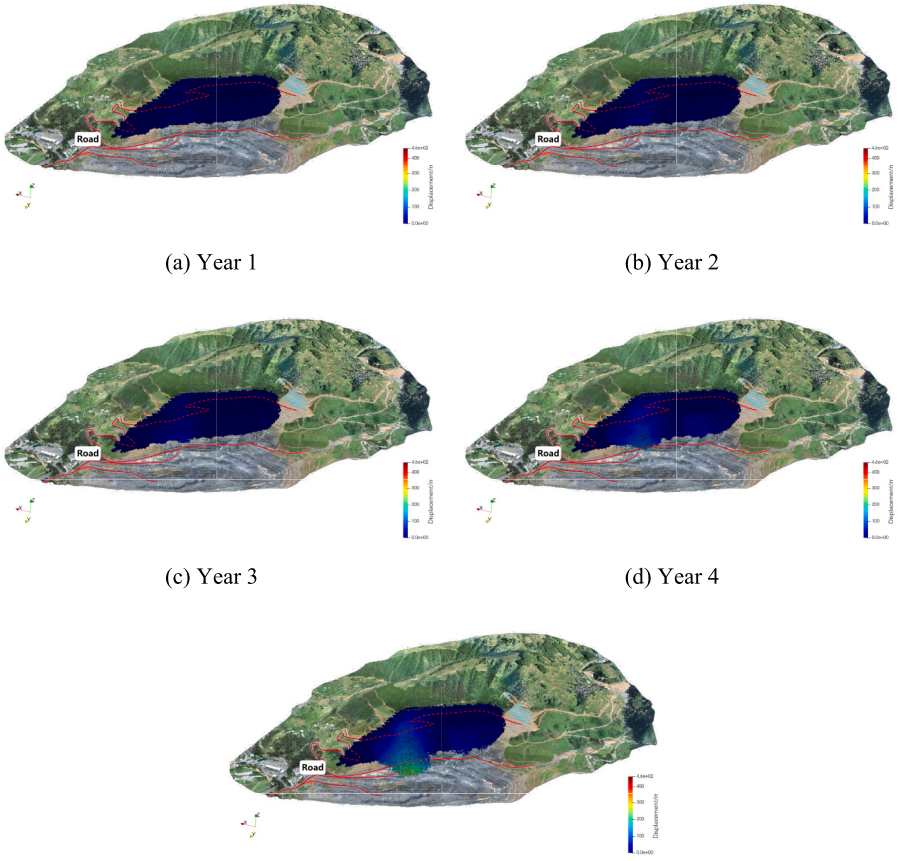


Fig. 13. Degradation Law of Strength Parameters for the Sliding Mass and Sliding Zone: Cohesion (Left) and Internal Friction Angle (Right).

results from the previous analysis are inherited by the next step, simulating the long-term damage to the Taihe landslide under the effect of strength degradation caused by alternating dry and rainy seasons over 10 years.

The code utilized in this study is based on the MPM 3D code open-sourced by Prof. Zhang from Tsinghua University. It could be integrated with pre-processing particle modeling and post-processing VTK reading/writing procedures, it formed the computational foundation for the extrapolation technology. The Taihe landslide example, aiming to simulate the landslide disaster over its 10-year evolution, underwent 10 stages of computation, taking a total of two weeks.

As the strength parameters of the landslide body degrade to the lowest level, the Taihe landslide exhibits a tendency toward catastrophic instability due to terrain changes caused by cumulative deformation. The landslide state shifts from local instability to complete catastrophic failure. The overall instability and accumulation of the landslide will eventually threaten the safety of the entire Taihe Mine production output. Without considering the mitigation of landslides: in the next five



(a) Year 1 (b) Year 2

(c) Year 3 (d) Year 4

(e) Year 5

Fig. 14. Simulation of the Effect and Damage Process of the Taihe Landslide over the Next 5 Years.

years, as the number of wet-dry cycles increases with the alternating dry and rainy seasons, displacement on the east side of the Taihe landslide continues to accumulate, while the west side remains relatively stable (Fig. 14). By the fifth year, the landslide body on the east side moves along the bedrock-deposit interface toward the free face, with the east-side collapse covering over 260 m; the deformation on the west side exceeds 63.4 m. In the next six to ten years, instability on both the east and west sides intensify, resulting in a complete landslide. As shown in Fig. 15, by the tenth year, the landslide deposits will cover over 460 m.

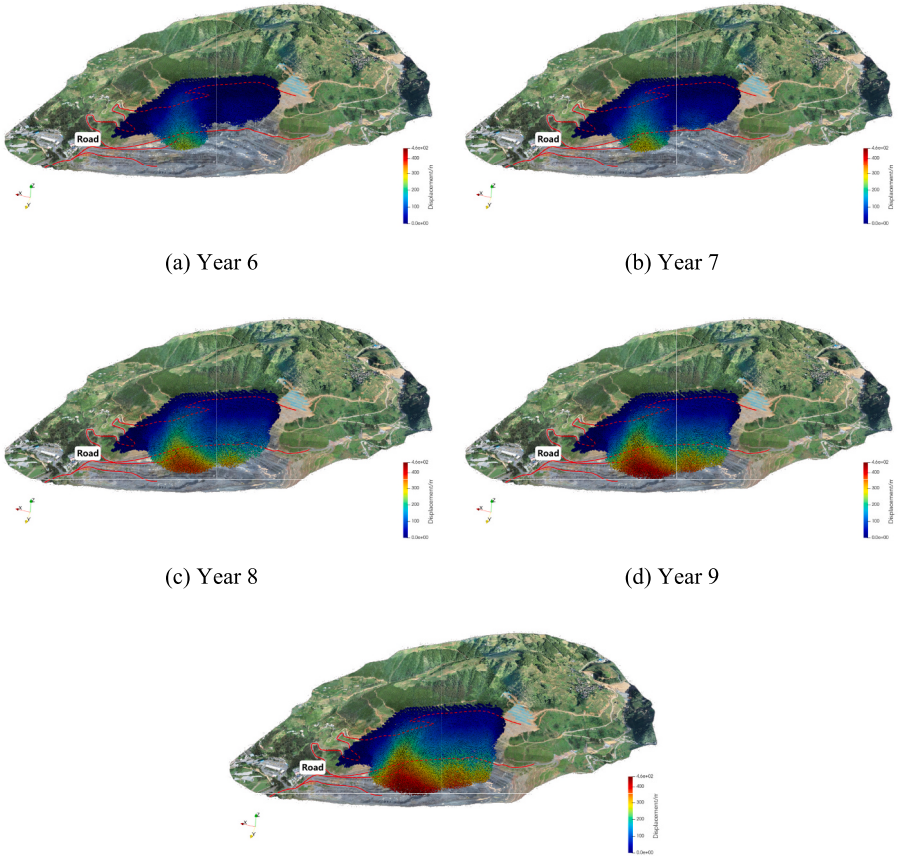
After 10 years of wet-dry cycles, the internal friction angle of the Taihe landslide degrades to 5°. The instability resulting from the degradation of the friction angle becomes particularly noticeable after five years. On the east side of the landslide, a more significant displacement occurs in the sixth year, based on the accumulated deformation from the first five years, while the west side collapses in the ninth year after eight years of cumulative deformation. In fact, the lower the strength, the more sensitive the landslide movement becomes to parameter degradation.

4. Conclusion

Taking the Taihe landslide as a case study, a digital simulation framework for geological disaster prediction was presented based on the MPM for large deformations. From a digital twin perspective, a digital twin model of geological disasters and their surrounding environments is established based on survey data. By integrating monitoring data to map the current state of the disaster prototype system and incorporating the time-dependent characteristics derived from the state evolution of geotechnical materials, the disaster state inversion and prediction simulation are performed using the twin model. This enables the twin model to characterize the actual state of the disaster, with a "multi-dimensional" time span, "multiple" physical information, and a "diverse" structural framework. It facilitates digital analysis and visualization of various static and dynamic physical data within geological disaster scenarios. The following key conclusions could be drawn:

(1) A digital simulation framework for geological disasters has been developed based on the MPM, where sensing data of the prototype system state and the twin model constitute the core data elements for digital simulation. This framework includes three steps: 1) Constructing a disaster-scenario twin model by extracting key elements from the disaster-prone scene, which serves as the data storage medium for digital simulation; 2) Performing state inversion of the disaster prototype system based on sensing data from the prototype disaster-prone scene, refining the selection of physico-mechanical parameters; 3) Updating the twin model driven by physico-mechanical models of geotechnical materials, and forecasting the future evolution of the disaster.

(2) Taking the Taihe landslide as a case, multi-source monitoring data are utilized to carry out the inversion of landslide states. By comparing the deformation direction, range, and local collapse position between the twin model and the landslide prototype, the input parameters of the twin model are constrained and optimized. Based on this, the long-term stability of the landslide is simulated by incorporating the strength evolution of geotechnical materials, and the future disaster evolution of the Taihe landslide over the next ten years is predicted. The Taihe landslide twin



(a) Year 6 (b) Year 7

(c) Year 8 (d) Year 9

(e) Year 10

Fig. 15. Simulation of the Effect and Damage Process of the Taihe Landslide over the Next 6–10 Years.

model thus reflects the time span element associated with its long-term service performance.

Nevertheless, the simulation-based extrapolation technology in this study has certain application limitations. At first, the discretization of phased time and the coarse-grained spatial discretization in the analysis of 10-year stability evolution are dictated by computational cost constraints. Secondly, the MPM-based technology is more applicable to colluvial and soil-landslides; the simulation of rock-landslides and collapses requires the development of new algorithms and models. Thirdly, the current technology offers an inadequately refined portrayal of landslide initiation, necessitating the future development of MPM algorithms incorporating rainfall and seismic triggers. Furthermore, given the uncertainty in landslide prototype data like the inference of geological lithology and the selection of material parameters, future work will delve into uncertainty analysis of prototype data and sensitivity analysis of model input parameters.

CRediT authorship contribution statement

Wenjie Du: Writing – original draft, Software, Formal analysis. **Xiaodong Fu:** Methodology, Investigation. **Qian Sheng:** Supervision, Methodology. **Jian Chen:** Writing – review & editing, Validation, Project administration. **Yongqiang Zhou:** Visualization. **Shaojie Zheng:** Writing – original draft.

Declaration of competing interest

None.

Acknowledgments

This study has received funding from the National Key R&D Program of China (2023YFC3081200), the National Natural Science Foundation of China (No. U21A20159; No. 52179117; No. 52409147), and the Key R&D Program of Yunnan Province (202303AA080010).

Data availability

Data will be made available on request.

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